

## SULINA, Romania

WMO: 153600

Lat: 45.1624N

Lon: 29.7266E

Elev: 13

StdP: 101.16

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.2       | -6.6 | -11.7                         | 1.4 | -8.3 | -9.5 | 1.7 | -5.7 | 20.2                  | 0.1  | 17.8 | -0.3 | 10.2                  | 0    | 0.631 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 4.9                    | 28.9            | 24.0 | 27.8 | 23.3 | 27.0 | 22.8 | 25.5                | 27.5 | 24.4 | 26.7 | 23.6 | 26.0 | 3.7                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.9                            | 19.9 | 26.9 | 23.7 | 18.6 | 26.1 | 22.8 | 17.5 | 25.4 | 78.3          | 27.5 | 74.0 | 26.7 | 70.8 | 26.0 | 29.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 16.3          | 14.1 | 12.2 | -12.0                      | 31.4 | 3.1                | 1.2 | -14.2                                              | 32.3 | -16.0      | 33.0 | -17.7      | 33.7 | -20.0      | 34.6 |     |     |
| (5)               |      |      | -12.3                      | 26.2 | 3.1                | 1.2 | -14.5                                              | 27.1 | -16.3      | 27.8 | -18.0      | 28.6 | -20.2      | 29.5 | (4) | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.4  | 0.9   | 1.9   | 5.4   | 10.7  | 16.9  | 21.3  | 23.7  | 23.6  | 19.0  | 13.5  | 8.4   | 3.2   | (6)  |
| (7)  |                                                                                  | DBStd       | 8.77  | 4.19  | 3.95  | 3.26  | 2.86  | 2.86  | 2.57  | 2.16  | 2.36  | 2.95  | 3.52  | 4.01  | 4.18  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 979   | 283   | 227   | 146   | 24    | 1     | 0     | 0     | 0     | 0     | 11    | 76    | 212   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2651  | 541   | 461   | 402   | 228   | 63    | 6     | 1     | 1     | 28    | 154   | 299   | 469   | (9)  |
| (10) |                                                                                  | CDD10.0     | 1861  | 0     | 0     | 2     | 46    | 213   | 340   | 425   | 420   | 269   | 118   | 26    | 1     | (10) |
| (11) |                                                                                  | CDD18.3     | 493   | 0     | 0     | 0     | 0     | 17    | 96    | 167   | 162   | 47    | 3     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 2230  | 0     | 0     | 0     | 0     | 12    | 291   | 921   | 943   | 63    | 1     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 258   | 0     | 0     | 0     | 0     | 1     | 23    | 104   | 128   | 2     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 6.1   | 6.6   | 6.7   | 7.0   | 6.2   | 5.6   | 5.5   | 5.1   | 5.2   | 5.8   | 6.2   | 6.4   | 6.9   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 276   | 18    | 18    | 16    | 17    | 25    | 30    | 23    | 27    | 28    | 19    | 23    | 24    | (15) |
| (16) |                                                                                  | PrecMax     | 487   | 74    | 84    | 64    | 72    | 142   | 120   | 90    | 129   | 126   | 114   | 95    | 114   | (16) |
| (17) |                                                                                  | PrecMin     | 109   | 1     | 2     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 2     | 0     | 0     | (17) |
| (18) |                                                                                  | PrecStd     | 72    | 14    | 16    | 13    | 12    | 25    | 22    | 20    | 27    | 27    | 18    | 20    | 22    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 9.9   | 10.8  | 15.1  | 19.6  | 24.7  | 28.8  | 29.9  | 30.7  | 26.3  | 22.3  | 17.0  | 13.1  | (19) |
| (20) |                                                                                  |             | MCWB  | 7.5   | 7.8   | 9.8   | 15.4  | 19.7  | 22.6  | 24.0  | 24.6  | 21.2  | 19.7  | 15.0  | 11.4  | (20) |
| (21) |                                                                                  | 2%          | DB    | 8.2   | 8.5   | 11.7  | 17.3  | 22.9  | 27.2  | 28.7  | 29.2  | 24.9  | 20.7  | 15.7  | 11.4  | (21) |
| (22) |                                                                                  |             | MCWB  | 7.0   | 6.7   | 8.8   | 14.3  | 19.1  | 22.3  | 23.9  | 24.5  | 21.0  | 18.5  | 14.4  | 10.1  | (22) |
| (23) |                                                                                  | 5%          | DB    | 7.1   | 7.3   | 10.1  | 15.6  | 21.7  | 26.0  | 27.8  | 28.0  | 23.9  | 19.5  | 14.6  | 10.0  | (23) |
| (24) |                                                                                  |             | MCWB  | 6.1   | 6.1   | 7.9   | 12.9  | 18.4  | 21.7  | 23.5  | 23.5  | 20.5  | 17.5  | 13.2  | 8.9   | (24) |
| (25) |                                                                                  | 10%         | DB    | 6.0   | 6.4   | 9.0   | 14.2  | 20.7  | 25.0  | 27.0  | 27.1  | 23.1  | 18.3  | 13.5  | 8.9   | (25) |
| (26) |                                                                                  |             | MCWB  | 5.3   | 5.5   | 7.5   | 12.0  | 17.8  | 21.1  | 23.1  | 22.8  | 20.0  | 16.2  | 12.4  | 7.9   | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 8.3   | 8.3   | 10.8  | 16.4  | 21.2  | 24.6  | 26.6  | 27.2  | 23.3  | 20.7  | 15.6  | 11.8  | (27) |
| (28) |                                                                                  |             | MCDWB | 8.9   | 9.8   | 13.4  | 18.8  | 23.4  | 26.8  | 28.2  | 28.8  | 24.5  | 21.7  | 16.4  | 12.6  | (28) |
| (29) |                                                                                  | 2%          | WB    | 7.2   | 7.1   | 9.3   | 14.7  | 19.9  | 23.5  | 25.5  | 25.9  | 22.4  | 19.0  | 14.5  | 10.4  | (29) |
| (30) |                                                                                  |             | MCDWB | 7.9   | 8.0   | 10.8  | 16.8  | 21.9  | 25.9  | 27.4  | 28.2  | 23.7  | 20.2  | 15.4  | 11.1  | (30) |
| (31) |                                                                                  | 5%          | WB    | 6.2   | 6.4   | 8.4   | 13.4  | 19.0  | 22.7  | 24.5  | 24.7  | 21.5  | 17.7  | 13.5  | 9.1   | (31) |
| (32) |                                                                                  |             | MCDWB | 6.9   | 7.1   | 9.6   | 15.0  | 21.0  | 25.1  | 26.7  | 26.9  | 23.0  | 19.2  | 14.3  | 9.8   | (32) |
| (33) |                                                                                  | 10%         | WB    | 5.4   | 5.6   | 7.8   | 12.4  | 18.2  | 21.8  | 23.7  | 23.7  | 20.7  | 16.4  | 12.6  | 8.0   | (33) |
| (34) |                                                                                  |             | MCDWB | 6.0   | 6.3   | 8.9   | 13.9  | 20.3  | 24.3  | 26.2  | 26.1  | 22.4  | 18.0  | 13.4  | 8.9   | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 4.3   | 4.0   | 3.8   | 3.5   | 3.8   | 4.5   | 4.9   | 4.5   | 4.2   | 4.0   | 4.4   | (35)  |      |
| (36) |                                                                                  |             | MCDBR | 4.9   | 5.3   | 5.9   | 5.2   | 5.3   | 5.8   | 5.6   | 5.8   | 5.1   | 4.8   | 4.2   | 5.0   | (36) |
| (37) |                                                                                  |             | MCWBR | 3.8   | 3.6   | 3.7   | 3.3   | 3.4   | 3.5   | 3.4   | 3.6   | 3.5   | 3.7   | 3.0   | 4.0   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 4.4   | 4.5   | 5.0   | 4.4   | 4.5   | 4.9   | 4.7   | 4.8   | 4.4   | 4.4   | 3.5   | 4.4   | (38) |
| (39) |                                                                                  |             | MCWBR | 3.8   | 3.4   | 3.4   | 3.1   | 3.4   | 3.4   | 3.4   | 3.8   | 3.6   | 3.9   | 2.9   | 3.8   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.352 | 0.380 | 0.408 | 0.437 | 0.416 | 0.414 | 0.432 | 0.439 | 0.413 | 0.391 | 0.368 | 0.351 | (40)  |      |
| (41) |                                                                                  | taud        | 2.319 | 2.253 | 2.209 | 2.166 | 2.315 | 2.351 | 2.351 | 2.311 | 2.280 | 2.317 | 2.340 | 2.335 | 2.329 | (41) |
| (42) |                                                                                  | Ebn at Noon | 736   | 783   | 815   | 823   | 855   | 856   | 835   | 813   | 802   | 763   | 710   | 693   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 81    | 104   | 124   | 141   | 125   | 121   | 125   | 124   | 111   | 95    | 80    | 74    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.16  | 1.94  | 3.33  | 4.90  | 6.24  | 6.67  | 6.62  | 5.89  | 4.24  | 2.60  | 1.37  | 1.00  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.20  | 0.27  | 0.33  | 0.42  | 0.39  | 0.37  | 0.32  | 0.31  | 0.34  | 0.23  | 0.20  | 0.16  | (45)  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) Station Only            |  | +0.52 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +78     | +61     |  |  |
| (47) Regional (15 neighbors) |  | +0.57 | N/A     | N/A    | +0.72   | N/A   | N/A   | N/A         | -132    | +111    | +77     |  |  |

Nomenclature: See separate page